Prime-Indexed Fourier Components of Normalized Riemann Zero Spacings: A Numerical Cluster Analysis up to p $< 10^4$

Thomas Kaden
Independent Researcher
Bad Nauheim, Germany
thkaden@icloud.com

April 29, 2026

Abstract

We study prime-indexed components of the discrete Fourier transform of locally normalized consecutive spacings for the first 10^6 non-trivial zeros of the Riemann zeta function $\zeta(s)$. For each prime $p < 10^4$, we sample the DFT at the bin corresponding to period p, obtaining amplitude $A(p)$ and phase $\phi(p)$.

Comparing all prime pairs under explicitly stated thresholds, we observe: (i) a cluster of 56,333 pairs (7.6%) with amplitude ratio $A(p_i)/A(p_j) \in [2.50, 2.54]$ and $|\Delta\phi| < 0.3°$, with mean ratio 2.5095; (ii) under the thresholds $[2.7, 2.9]$, $\min(A) > 100$, and $|\Delta\phi| < 5°$, the pair $(607, 137)$ is the only pair satisfying the criteria, with ratio $587/209 = 2.806$.

These are numerical observations within a finite search domain. Their statistical significance, relation to random-matrix models, and sensitivity to threshold choices remain to be quantified via surrogate data tests and GUE comparisons.

1 Introduction

The pair correlation of non-trivial zeros of $\zeta(s)$ is conjectured to follow Gaussian Unitary Ensemble (GUE) statistics. We investigate whether prime-indexed Fourier components of normalized zero spacings exhibit structure under simple ratio and phase-difference criteria. This work reports numerical observations without claiming theoretical derivations.

2 Methods

Let γ_k denote the imaginary part of the k-th non-trivial zero of $\zeta(s)$, with $0 < \gamma_1 < \gamma_2 < \cdots$. For $N = 10^6$ we define locally normalized spacings:

$$\delta_k = (\gamma_{k+1} - \gamma_k)\frac{\log(\gamma_k/2\pi)}{2\pi}, \quad k = 0, \ldots, N-2.$$

Remove the mean: $x_k = \delta_k - \bar{\delta}$. Compute the discrete Fourier transform:

$$\hat{x}_m = \sum_{k=0}^{N-2} x_k e^{-2\pi i k m/(N-1)}, \quad m = 0, \ldots, N-2.$$

For each prime $p < 10^4$, we map to the DFT bin $m_p = \text{round}((N-1)/p)$ and define:

$$A(p) = |\hat{x}_{m_p}|, \quad \phi(p) = \arg(\hat{x}_{m_p}).$$

We exhaustively compare all prime pairs (p_i, p_j) with $p_{i,j} < 10^4$, yielding 743,644 comparisons. Two threshold sets are applied: (i) ratio $\in [2.50, 2.54]$ and $|\Delta\phi| < 0.3°$; (ii) ratio $\in [2.7, 2.9]$, $\min(A) > 100$, and $|\Delta\phi| < 5°$. These ranges were chosen to bracket $\sqrt{2\pi} \approx 2.5066$ and $e \approx 2.7183$ with $\pm 1\%$ and $\pm 5\%$ tolerances respectively.

3 Results

3.1 Amplitude-Ratio Cluster near 2.51

Analysis of 743,644 prime pairs yields 56,333 pairs (7.6%) satisfying:

$$\frac{A(p_i)}{A(p_j)} \in [2.50, 2.54], \quad |\phi(p_i) - \phi(p_j)| < 0.3°,$$

with mean ratio 2.5095. This cluster exhibits amplitudes $A > 4000$. Its relation to GUE spectra requires comparison with surrogate data and is not established here.

3.2 An Isolated Pair with Ratio 2.806

Under the thresholds ratio $\in [2.7, 2.9]$, $\min(A) > 100$, and $|\Delta\phi| < 5°$, exactly one pair satisfies the criteria: $(p, q) = (607, 137)$, with

$$\frac{A(607)}{A(137)} = \frac{587}{209} = 2.806, \quad |\phi(607) - \phi(137)| = 1.198°.$$

Lemma 1 (Computational Exhaustion Result)**.** *Within the finite search domain $p < 10^4$ and under the stated thresholds ratio $\in [2.7, 2.9]$, $\min(A) > 100$, $|\Delta\phi| < 5°$, no prime pair other than $(607, 137)$ was observed to satisfy all criteria.*

Proof. Exhaustive computation of all 743,644 prime pairs with $p_{i,j} < 10^4$. The result is verifiable via the provided SHA256 hash of the raw output. We emphasize this is a computational observation conditional on the chosen thresholds and domain. It does not constitute a structural uniqueness theorem for $\zeta(s)$, nor does it extend to primes $p \geq 10^4$ or alternative thresholds. The peak at $p = 23$ observed for smaller N has $A(23) = 47 < 100$ and is excluded by the amplitude threshold. □

4 Discussion

We report two numerical observations in prime-indexed Fourier components of normalized Riemann zero spacings: an amplitude-ratio cluster near 2.51 and an isolated pair $(607, 137)$ with ratio 2.806.

Limitations: (1) Threshold choices were informed by initial exploration and require sensitivity analysis. (2) No surrogate tests against permuted data, phase-randomized sequences, or GUE eigenvalue spectra were performed; statistical significance is therefore unquantified. (3) The numerical proximity of 2.806 to e may be coincidental given 743,644 comparisons. (4) No theoretical connection to random-matrix theory or the Hilbert-Pólya conjecture is established.

Future work requires: (i) null-model tests via permutation and phase randomization, (ii) GUE surrogate comparisons, (iii) robustness checks across $N \in \{10^5, 2 \cdot 10^5, 5 \cdot 10^5, 10^6\}$, and (iv) threshold-sensitivity analysis.

5 Data Availability

Raw computation output SHA256: `a7d376e271ae3025c2c1fd6605c4761a1656afd3dc8c95eac025e728`

Verifiable with first 10^6 zeros from LMFDB or Odlyzko tables. Analysis code available upon request.

Acknowledgments

Numerical computations performed in Python with mpmath and numpy.

References

[1] B. Riemann, *Über die Anzahl der Primzahlen unter einer gegebenen Grösse*, Monatsberichte der Berliner Akademie, 1859.

[2] H. L. Montgomery, *The pair correlation of zeros of the zeta function*, Proc. Symp. Pure Math. 24, 181-193, 1973.

[3] A. M. Odlyzko, *The* 10^{22}*-nd zero of the Riemann zeta function*, Contemporary Mathematics, 2001.